01 / Explain the system

Follow the system from light to a saved observation. Draw the camera hardware, Pi camera stack, Python wrapper, perception function, and report. Add the saved-image branch. Mark which parts own hardware and which can run without it.

What crosses each boundary? Label arrays, dictionaries, and files.

Where does RGB become BGR? Where would a learned detector fit?

02 / Read Python with a purpose

Use the complete source, not memorized definitions. Choose a function, class instance, dictionary lookup, array slice, condition, and loop. Explain what each contributes to one observation.

Trace one call to analyze: arguments, local values, return values.

Explain self.device and the with-block cleanup in your own words.

03 / Predict, change, observe

Choose one saved original image. Change exactly one setting. Keep the input fixed so your comparison is about the code. Record the exact command and output-folder name so you can repeat the experiment.

Prediction and changed setting:

Observed area / found / image region; explanation and next test:

04 / Investigate a failure

A rule can run correctly and still miss your intended target. Try a similar-colored distractor, a background, shadows, or two touching objects. Distinguish a bad input or runtime error from a valid but unhelpful observation.

Describe or sketch the scene and the actual implemented rule.

What evidence suggests the cause? What single change would test it?

05 / Own your modification

Add area_fraction, or try the optional selection policy. Explain every value you use and the meaning of the output. A fraction of image pixels is not a confidence score or a distance.

Write the code change and its purpose. Which module should own it?

Record an input that should change and one that should stay the same.

06 / Teach back and next step

Demonstrate a real-camera success, a controlled comparison, and one limitation. Explain why this color rule does not know object categories. Choose one next module and define the evidence it should return.

Three ideas I can explain using my own code:

One uncertainty, one next project, and the first test I would run: